ANALYSIS REPORT

Harmonic-Wise Paper Reimplementation Pipeline

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This guide explains the repository-owned harmonic-wise pipeline that was added to reproduce the offline logic of `reference/RCIM_ML-compensation.pdf`.

This topic is not the same as the repository's direct-TE training families.

It belongs to the paper-faithful reproduction branch, which the backlog now calls `RCIM Model-Bank Reproduction`.

The purpose of this branch is clear:

- recreate the paper's harmonic-wise modeling logic inside this repository;
- obtain an offline comparison protocol that is genuinely paper-aligned;
- prepare the bridge toward later online compensation work;
- keep the result inspectable for colleagues, reviewers, and future

  deployment work. At the current repository state, this guide covers the offline branch only. The online compensation loop and the final `Table 9` style benchmark are still future work.

Guide Export Folders

This topic separates its guide-local material into:

- `assets/`

  diagrams and `NotebookLM` source packages.
- `English/`

  imported final English `NotebookLM` exports for the concept and project tracks.
- `Italiano/`

  imported final Italian `NotebookLM` exports for the concept and project tracks. The imported export naming follows the repository rule that each filename must explicitly declare:
- guide name;
- track (`Concept` or `Project`);
- artifact type (`Guide` , `Presentation` , `Video Overview` , `Mind Map` ,

  `Infographic`). Language is intentionally expressed by the parent folder name, so it is not repeated again inside each imported filename.

Why This Pipeline Exists

The repository had already trained several strong direct-TE models.

Those models predict the TE curve directly from angular position plus operating conditions.

That is useful, but it is not how the reference paper is structured.

The paper works in a different way:

1. represent each TE curve through selected harmonics;
2. predict the harmonic quantities from operating conditions;
3. reconstruct the TE curve from the predicted harmonic stack;
4. later apply that reconstructed TE information to compensation logic.

   So this pipeline exists because the repository needed a branch that answers a different question: can the paper's harmonic-wise logic be reproduced here, with repository-owned code, artifacts, and reports?

Track Placement

The repository now keeps two comparison tracks:

- `RCIM Model-Bank Reproduction`

  paper-faithful harmonic-wise reproduction.
- `TE Curve Verification Pipeline`

  repository direct-TE comparison under a common offline TE-curve evaluator. This guide is only about `RCIM Model-Bank Reproduction`.

RCIM Model-Bank Reproduction Completion Rule

The canonical `RCIM Model-Bank Reproduction` completion rule is not a single campaign winner.

`RCIM Model-Bank Reproduction` is now closed as the faithful full-dataset model-bank reproduction. Closure means the repository has run both forward and backward paper-faithful grid-search campaigns, refreshed accepted archives, and repopulated Tables `2`, `3`, `4`, and `5` with inspectable per-target and per-family status.

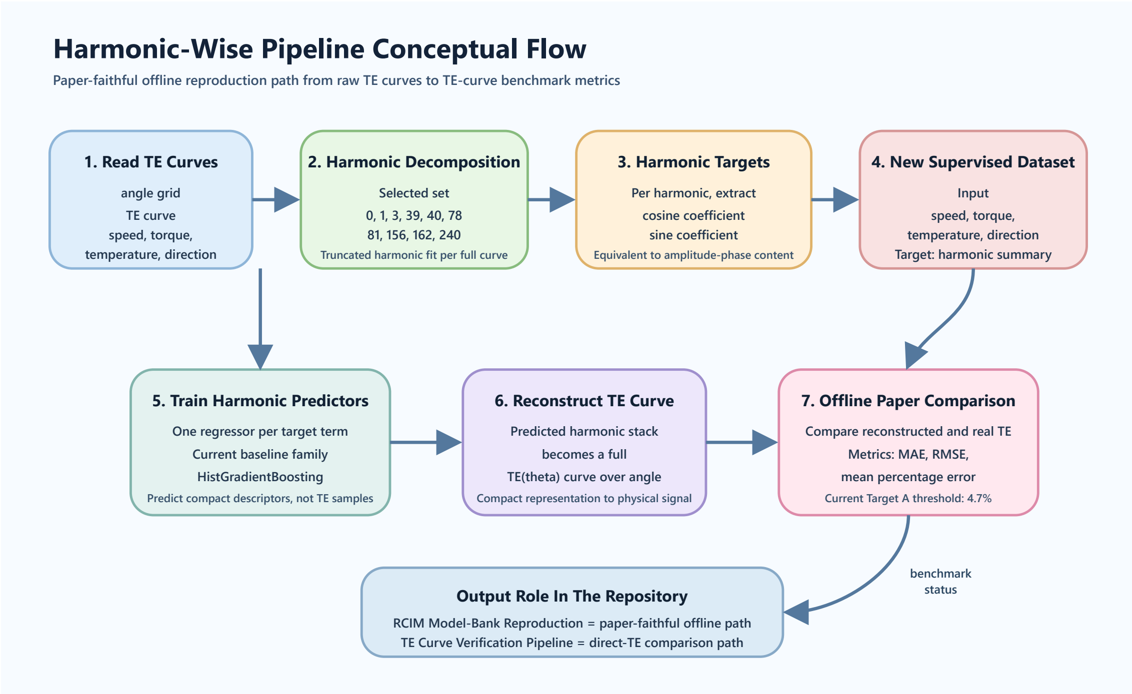
That means

- the primary `RCIM Model-Bank Reproduction` readout is the exact-paper table-replication report;
- the harmonic-wise TE-curve evaluator is still useful, but only as support

  evidence;
- all-green benchmark coloring is not required for the faithful RCIM Model-Bank Reproduction

  closure; yellow and red cells remain evidence of the numerical gap under the recovered original protocol.

End-To-End Flow



Harmonic-wise pipeline conceptual flow

The pipeline starts from full TE curves, converts them into a harmonic summary, predicts those harmonic terms from operating conditions, and then reconstructs the TE curve for evaluation.

That means the model does not learn point-wise TE samples directly.

It learns a compact representation of the whole curve.

Stage 1. Read Curves And Operating Conditions

Each dataset item contains

- an angular grid;
- a TE curve on that angular grid;
- operating conditions such as `speed`, `torque`, `temperature`, and `direction`. In the direct-TE branch, the angle itself is part of the model input. In this harmonic-wise branch, angle is not the final learning input. Instead, the full angle-resolved TE curve is treated as a signal to summarize.

Stage 2. Decompose The TE Curve Into Selected Harmonics

The pipeline uses a selected harmonic set aligned with the paper-oriented comparison scope:

- `0`
- `1`
- `3`
- `39`
- `40`
- `78`
- `81`
- `156`
- `162`
- `240`

  These harmonics were chosen because they capture the practical periodic structure emphasized by the paper and by the repository's own benchmark notes. The decomposition stage fits a truncated harmonic representation of the TE curve. Conceptually, the curve is written as: `TE(theta) ~= a0 + sum_k [c_k cos(k theta) + s_k sin(k theta)]` where:

- `a0` is the offset term;
- `c_k` is the cosine coefficient for harmonic `k`;
- `s_k` is the sine coefficient for harmonic `k`.

Stage 3. Convert The Curve Into Harmonic Targets

For each selected harmonic, the current implementation stores the signal in coefficient form:

- cosine coefficient;
- sine coefficient.

  This form is convenient for stable regression and reconstruction. It is also directly connected to the paper quantities:

- amplitude `A_k`;
- phase `phi_k`.

  The relationship is:

- `A_k = sqrt(c_k^2 + s_k^2)`
- `phi_k = atan2(-s_k, c_k)` in the chosen convention

  So even though the repository currently predicts coefficient pairs, those coefficients still represent the same harmonic content that can be expressed as `A_k` and `phi_k`.

Stage 4. Build A New Supervised Dataset

This is the key conceptual shift.

The problem is no longer

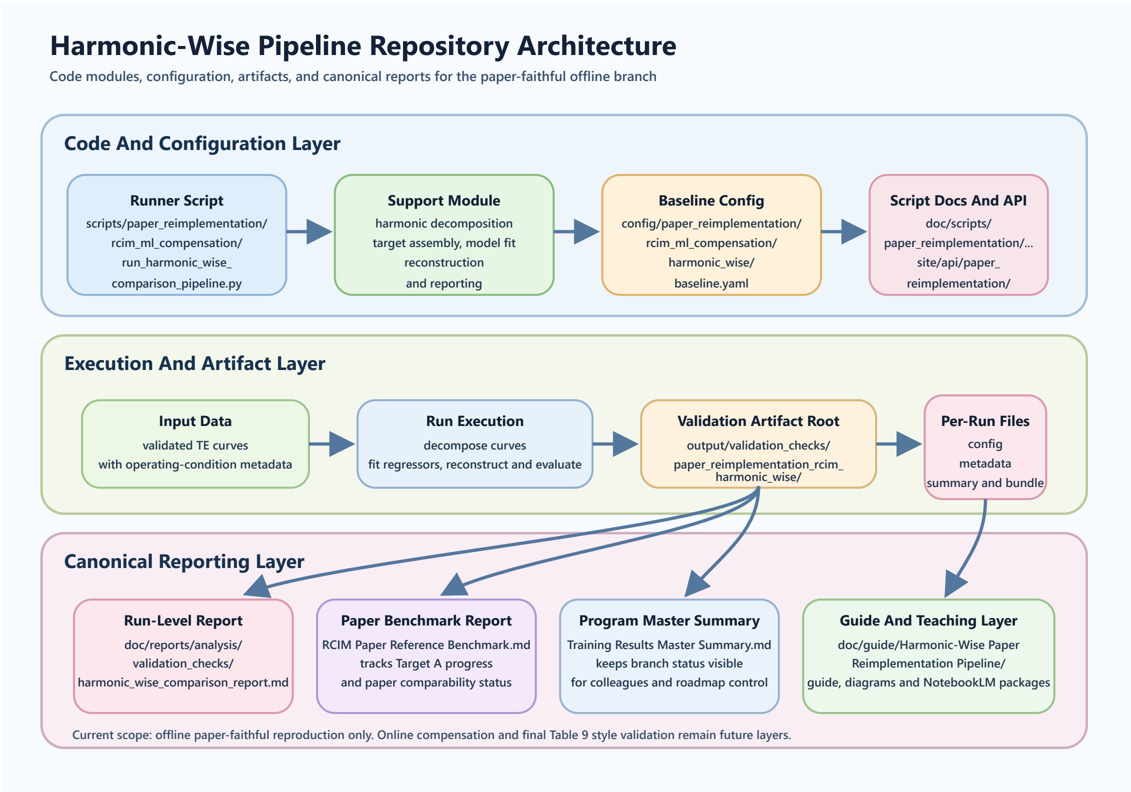
- input = `angle + operating conditions`
- target = `TE(angle)`

  The problem becomes:

- input = `speed + torque + temperature + direction`
- target = harmonic summary of the whole TE curve

  So each supervised sample corresponds to one operating-condition point and one whole-curve harmonic representation. That is why the pipeline is called harmonic-wise. It predicts the ingredients of the curve, not the point-wise curve samples.

Stage 5. Train Predictors On Harmonic Targets



Harmonic-wise repository architecture

The current repository baseline uses

- one regressor per harmonic target term;
- model family: `HistGradientBoosting`.

  That means the pipeline fits many small tabular regressors rather than one large end-to-end point-wise neural model. This is already directionally aligned with the paper, where tree and boosting families dominate the deployable harmonic stack. The baseline script is:

- `scripts/paper_reimplementation/rcim_ml_compensation/run_harmonic_wise_comparison_pipeline.py`

  The main support module is:

- `scripts/paper_reimplementation/rcim_ml_compensation/harmonic_wise_support.py`

  The current baseline configuration is:

- `config/paper_reimplementation/rcim_ml_compensation/harmonic_wise/baseline.yaml`

Stage 6. Reconstruct The TE Curve

Once the model has predicted the harmonic coefficients for a new operating point, the pipeline rebuilds the full TE curve over angle.

This is the bridge between

- compact harmonic prediction;
- physical TE behavior over the full angular cycle.

  Without reconstruction, the pipeline would only output abstract coefficients. With reconstruction, it becomes comparable to the paper's offline TE-curve validation logic.

Stage 7. Evaluate The Reconstructed Curve

The reconstructed TE curve is compared against the real TE curve on held-out validation and test splits.

The main metrics are

- `MAE`
- `RMSE`
- mean percentage error

  The mean percentage error over the full TE curve is now a supporting paper-comparable metric for the offline phase. The first repository-owned baseline established that protocol and produced:

- validation mean percentage error: `9.474%`
- test mean percentage error: `9.403%`

  The current offline paper target is:

- `<= 4.7%`

  So the comparison protocol now exists, but `Target A` is not closed yet. RCIM Model-Bank Reproduction is closed separately from `Target A`: the model-bank reproduction surface has been populated and archived, while the later TE-curve and online compensation branches still need their own validation.

Why The First Baseline Still Matters

Even though the first baseline does not yet match the paper threshold, it already proved something important.

The truncation-only oracle with the same harmonic set reaches a much lower error than the learned predictor.

That means

- the selected harmonic set is expressive enough;
- the bottleneck is the operating-condition predictor;
- the next improvement work should focus on the predictor and target design,

  not on abandoning the harmonic representation itself.

Offline Playback Probes

The current pipeline also includes offline `Robot` and `Cycloidal` style motion-profile playback probes.

These are not yet the final online compensation benchmark.

They are repository-owned preparation steps that

- exercise the predicted harmonic stack on profile-like conditions;
- prepare the path toward later online evaluation;
- keep the eventual `Table 9` target visible from the beginning.

What This Pipeline Is Not

This pipeline is not

- the direct-TE training branch;
- a finished online compensation system;
- a TwinCAT deployment package;
- the final `Table 9` benchmark.

  It is an intermediate but essential branch between completed `Wave 1` and the later online compensation path.

Current Repository Outputs

The current validation artifacts live under

- `output/validation_checks/paper_reimplementation_rcim_harmonic_wise/forward/`

  The current human-readable validation report is generated under:

- `doc/reports/analysis/validation_checks/`

  The canonical comparison reports that summarize the branch are:

- `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`
- `doc/reports/analysis/validation_checks/track1/exact_paper/forward/shared/2026-04-12-17-00-28_paper_reimplementation_rcim_exact_model_bank_rcim_exact_paper_model_bank_exact_paper_validation_tables_3_4_5_6_exact_paper_model_bank_report.md`

Practical Reading Map

If you want to understand the branch quickly, read in this order:

1. this guide;
2. `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md` ;
3. `doc/reports/analysis/validation_checks/track1/harmonic_wise/...harmonic_wise_comparison_report.md` ;
4. the pipeline script and support module.

Summary

The harmonic-wise paper-reimplementation pipeline changes the learning problem from:

- point-wise TE prediction

  to:

- operating-condition-to-harmonic prediction

  and then reconstructs the TE curve for evaluation. That is exactly why it matters. It gives the repository a paper-faithful offline benchmark path instead of only a direct-TE leaderboard. The important operational nuance is that the harmonic-wise support branch and the exact-paper table-replication branch must not be collapsed into one winner-centric story. For `RCIM Model-Bank Reproduction`, the table-replication branch is primary.

Related Reading

- `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`
- `doc/guide/project_usage_guide.md`